Functions: Modulus & Inverse

Section 1: Modulus Equations & Linear Graphs

1 Solve the modulus equation $|3x - 5| = 2x + 1$ for all real solutions.

a. $x = 6$ and $x = 0.8$

b. $x = 6$ only

c. $x = 4$ and $x = -2$

d. $x = 0.8$ only

2 Solve the modulus equation $|2x - 7| = |x + 2|$.

a. $x = 9$ and $x = 5/3$

b. $x = 9$ only

c. $x = 5$ and $x = -3$

d. $x = 3$ and $x = -1$

3 Find the coordinates of the vertex of the graph $y = |2x - 6| + 4$.

a. $(3, 4)$

b. $(-3, 4)$

c. $(3, -4)$

d. $(6, 4)$

Section 2: Modulus Inequalities

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4 Solve the inequality $|2x - 5| < 7$.

a. $-1 < x < 6$

b. $x < -1$ or $x > 6$

c. $-6 < x < 1$

d. $1 < x < 6$

5 Solve the inequality $|3x + 1| \geq |x - 4|$.

a. $x \leq -5/2$ or $x \geq 3/4$

b. $-5/2 \leq x \leq 3/4$

c. $x \leq -1$ or $x \geq 4$

d. $x \geq 3/4$

6 Find the set of values of x satisfying $|x - 2| > 2x - 1$.

a. $x < 1$

b. $x > 1$

c. $-1 < x < 1$

d. $x < 3$

Section 3: Composite and Inverse Functions in A2

7 The function $f(x) = 2 + \ln(x - 3)$ is defined for $x > 3$. Find the inverse function $f^{-1}(x)$ and state its domain.

a. $f^{-1}(x) = 3 + e^{x-2}$, domain $x \in \mathbb{R}$

b. $f^{-1}(x) = 3 - e^{x-2}$, domain $x > 0$

c. $f^{-1}(x) = 2 + e^{x-3}$, domain $x \in \mathbb{R}$

d. $f^{-1}(x) = \ln(x - 2) + 3$, domain $x > 2$

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8 Given $f(x) = e^{2x}$ for $x \in \mathbb{R}$ and $g(x) = \dfrac{1}{x}$ for $x > 0$, solve the equation $fg(x) = e^4$.

a. $x = 1/2$

b. $x = 2$

c. $x = 1/4$

d. $x = 4$

9 A function $h(x) = x^2 - 4x + 1$ is defined on the domain $x \leq k$. Find the greatest value of k for which h is one-to-one.

a. $k = 2$

b. $k = 4$

c. $k = 0$

d. $k = 1$

Section 4: Transformations & Analytical Modeling

10 Describe the transformation mapping the graph of $y = \ln(x)$ onto the graph of $y = \ln(3x - 6)$.

a. Horizontal stretch scale factor $1/3$ followed by translation $[2, 0]$

b. Translation $[6, 0]$ followed by horizontal stretch scale factor 3

c. Horizontal stretch scale factor 3 followed by translation $[-2, 0]$

d. Vertical stretch scale factor 3 followed by translation $[2, 0]$

11 Find the number of real roots of the equation $|2x - 3| = x^2 - 2$.

a. 2 roots

b. 4 roots

c. 0 roots

d. 1 root

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12 The graph of $y = f(x)$ passes through $(2, 5)$. What are the coordinates of the corresponding point on $y = 3 - 2f(x + 1)$?

a. $(1, -7)$

b. $(3, -7)$

c. $(1, 13)$

d. $(3, 7)$

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